



The Future Pricing

Consumption vs Investment Assets

- Investment assets are assets held by significant numbers of people purely for investment purposes
- Consumption assets are assets held primarily for consumption (Examples: copper, oil)

Short Selling

- Short selling involves selling securities you do not own
- Your broker borrows the securities from another client and sells them in the market in the usual way

Short Selling

- At some stage you must buy the securities so they can be replaced in the account of the client
- You must pay dividends and other benefits the owner of the securities receives
- There may be a small fee for borrowing the securities



Step-by-step process of Short Selling

Stock

HCL Technologies

Trade Type

Short (sell first and buy later)

Trade Type

Short (sell first and buy later)

Trade Duration

Intra day

Short Price

Rs.1990/-

Number of shares

50

Target Price

Rs.1950/-

% Profit Expected

2.0%

Stoploss

Rs.2000/-

Risk

Rs.10/-

Reward

Rs.40/-

Scenario 1 – The stock price hits the target of Rs.1950/-

In this case the stock has moved as per the expectation. The stock has fallen from Rs.1990/- to Rs.1950/-. Since the target has been achieved, the trader is expected to close the position. As we know in a short position the trader is required to –

1. First sell @ Rs.1990/- and
2. Later buy @ Rs.1950/-

In the whole process, the trader would have made a profit equal to the differential between the selling and buying price – i.e. Rs.40/- (1990 – 1950).

If you look at it from another angle (i.e. the usual buy first and sell later angle), this is as good as buying at Rs.1950 and selling at Rs.1990. It is just that the trader has reversed the transaction order by selling first and buying later.

Scenario 2 – The stock price increases to Rs.2000/-

In this case the stock has gone higher than the short price of Rs.1990/-. Recollect when you short, for you to profit the stock needs to decline in price. If the stock price goes up instead then there would be a loss. In this case the stock has gone up, hence there would be a loss –

1. The trader shorted @ Rs.1990/-. After shorting, the stock went up as opposed to the trader's expectation
2. The stock hits Rs.2000/- and triggers the stoploss. To prevent further losses, the trader will have to close the position by buying the stock back.

In the whole process the trader would have suffered a loss of Rs.10/- ($2000 - 1990$). If you look at it from the regular buy first sell later angle – this transaction is as good as buying at Rs.2000/- and selling at Rs.1990/ , and again if we reverse the order it would be sell first and buy later.

Notation for Valuing Futures and Forward Contracts

Assumptions:

- No transaction cost
- Unlimited borrowing and lending at risk free rate
- Avail Arbitrage opportunity

S_0 : Spot price today

F_0 : Futures or forward price today

T : Time until delivery date

r : Risk-free interest rate for maturity T

An Arbitrage Opportunity?

- Suppose that:
 - The spot price of a non-dividend-paying stock is \$40
 - The 3-month forward price is \$43
 - The 3-month risk free interest rate is 5% per annum (Continuously compounding)
- Is there an arbitrage opportunity?

Another Arbitrage Opportunity?

- Suppose that:
 - The spot price of non-dividend-paying stock is \$40
 - The 3-month forward price is US\$39
 - The 3-month interest rate is 5% per annum (continuously compounded)
- Is there an arbitrage opportunity?

Forward Price = \$43

Action now:

Borrow \$40 at 5% for 3 months

Buy one unit of asset

Enter into forward contract to sell
asset in 3 months for \$43

Action in 3 months:

Sell asset for \$43

Use \$40.50 to repay loan with interest

Profit realized = \$2.50

Forward Price = \$39

Action now:

Short 1 unit of asset to realize \$40

Invest \$40 at 5% for 3 months

Enter into a forward contract to buy
asset in 3 months for \$39

Action in 3 months:

Buy asset for \$39

Close short position

Receive \$40.50 from investment

Profit realized = \$1.50

The Forward Price

If the spot price of an investment asset is S_0 and the futures price for a contract deliverable in T years is F_0 , then

$$F_0 = S_0 e^{rT}$$

where r is the T -year risk-free rate of interest.

In our examples, $S_0 = 40$, $T = 0.25$, and $r = 0.05$ so that

$$F_0 = 40e^{0.05 \times 0.25} = 40.50$$

When an Investment Asset Provides a Known Income

$$F_0 = (S_0 - I)e^{rT}$$

where I is the present value of the income during life of forward contract

Example

- Consider a coupon bearing bond of current price Rs.900. Suppose the forward contract matures at 9 months. Assume that the coupon payment of Rs.40 is expected to receive after 4 months. The 4-month and 9-month risk free rates are 3% and 4% per annum. Determine the arbitrage opportunity if (i) forward price is Rs.910 and (ii) Rs. 870

Forward price = \$910

Action now:

Borrow \$900: \$39.60 for 4 months
and \$860.40 for 9 months

Buy 1 unit of asset

Enter into forward contract to sell
asset in 9 months for \$910

Action in 4 months:

Receive \$40 of income on asset
Use \$40 to repay first loan
with interest

Action in 9 months:

Sell asset for \$910
Use \$886.60 to repay second loan
with interest

Profit realized = \$23.40

Forward price = \$870

Action now:

Short 1 unit of asset to realize \$900

Invest \$39.60 for 4 months
and \$860.40 for 9 months

Enter into a forward contract to buy
asset in 9 months for \$870

Action in 4 months:

Receive \$40 from 4-month investment
Pay income of \$40 on asset

Action in 9 months:

Receive \$886.60 from 9-month investment
Buy asset for \$870
Close out short position

Profit realized = \$16.60

Example:

- Consider a 10 months forward contract on a stock when the stock price Rs. 50. Assume that the risk-free rate of interest is 8% per annum (continuously compounded). Also, the stock is expected to give dividend Rs. 0.75 per share after 3 months, 6 months, and 9 months. Determine the forward price.


$$I = 0.75e^{-0.08 \times 3/12} + 0.75e^{-0.08 \times 6/12} + 0.75e^{-0.08 \times 9/12} = 2.162$$

The variable T is 10 months, so that the forward price, F_0 , from equation (5.2), is given by

$$F_0 = (50 - 2.162)e^{0.08 \times 10/12} = \$51.14$$

When an Investment Asset Provides a Known Yield

$$F_0 = S_0 e^{(r-q)T}$$

where q is the average yield during the life of the contract (expressed with continuous compounding)

Example:

A 6-month forward contract on an asset that is expected to provide income equals to 4% per annum with semi-annual compounding. The spot price of the asset is Rs.25. The risk-free rate of interest is 10% per annum with continuous compounding. Determine the forward price.


$$R_c = m \ln \left(1 + \frac{R_m}{m} \right)$$

$$R_m = m(e^{R_c/m} - 1)$$

$$F_0 = 25e^{(0.10-0.0396) \times 0.5} = \$25.77$$

Home Assignment

ITC futures with expiration due on 28 May 2020 quoted at Rs.175.50 on 30 April 2020, while underlying was Rs.181.05. On the same day, one-month MIBOR was 4.96%. Calculate fair value of futures. You may ignore dividend since ITC would not pay any interim dividend in May. Is there any opportunity for earning arbitrage profit?

Valuing a Forward Contract

- A forward contract is worth zero when it is first negotiated
- Later it may have a positive or negative value
- Suppose that K is the delivery price and F_0 is the forward price for a contract that would be negotiated today

Valuing a Forward Contract

- By considering the difference between a contract with delivery price K and a contract with delivery price F_0 we can deduce that:
 - the value of a long forward contract is
$$(F_0 - K)e^{-rT}$$
 - the value of a short forward contract is
$$(K - F_0)e^{-rT}$$

Example

- A long forward contract on a non-dividend paying stock was entered into sometime ago. It has 6 months to maturity. The risk free rate of interest is 10% per annum (Continuously compounding), the current stock price is Rs.25 and delivery price is Rs.24. Determine the value of forward contract

Solution

$$F_0 = 25e^{0.1 \times 0.5} = \$26.28$$

$$f = (26.28 - 24)e^{-0.1 \times 0.5} = \$2.17$$

Stock Index

- Can be viewed as an investment asset paying a dividend yield
- The futures price and spot price relationship is therefore

$$F_0 = S_0 e^{(r-q)T}$$

where q is the average dividend yield on the portfolio represented by the index during life of contract

Example

- A stock index currently stands at Rs. 11250.55. The risk-free rate is 3.69% per annum and dividend yield on the index is 1.42% per annum. What is the forward price?

Example

- Assume that the risk-free rate 9% per annum and the dividend yield on the stock s varies throughout the year. In February, May, August and November the dividends are paid at 5% per annum. And, in the remaining months, it is paid at the rate of 2% per annum. Suppose that the value of the index on July 31 is Rs. 1300. What is the future price of a contract deliverable on December 31, of the same year?

The futures contract lasts for five months. The dividend yield is 2% for three of the months and 5% for two of the months. The average dividend yield is therefore

$$\frac{1}{5}(3 \times 2 + 2 \times 5) = 3.2\%$$

The futures price is therefore

$$1300e^{(0.04-0.032) \times 0.4167} = 1304.34$$

or \$1304.34.

Index Arbitrage

- When $F_0 > S_0 e^{(r-q)T}$ an arbitrageur buys the stocks underlying the index and sells futures
- When $F_0 < S_0 e^{(r-q)T}$ an arbitrageur buys futures and shorts or sells the stocks underlying the index

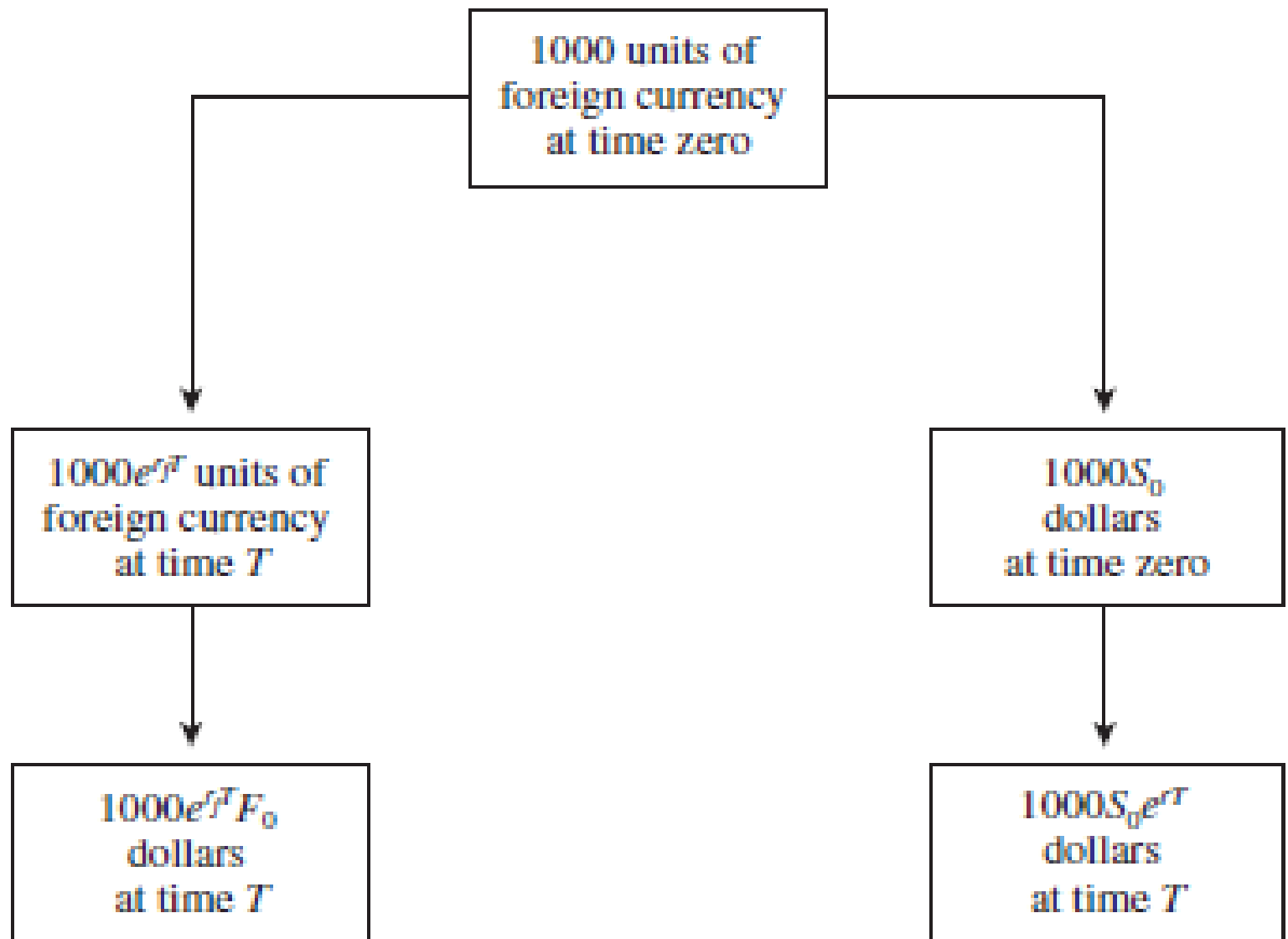
Index Arbitrage

- Index arbitrage involves simultaneous trades in futures and many different stocks
- Very often a computer is used to generate the trades
- Occasionally simultaneous trades are not possible and the theoretical no-arbitrage relationship between F_0 and S_0 does not hold

Futures and Forwards on Currencies

- A foreign currency is analogous to a security providing a yield
- The yield is the foreign risk-free interest rate
- It follows that if r_f is the foreign risk-free interest rate

$$F_0 = S_0 e^{(r - r_f)T}$$



Example

- Suppose that the 2-year interest rates in Australia and the United States are 3% and 1%, respectively, and the spot exchange rate is 0.9800 USD per AUD. Determine the forward price. Is there any arbitrage opportunity If the forward exchange rate is 0.93?

1. Borrow 1,000 AUD at 3% per annum for 2 years, convert to 980 USD and invest the USD at 1% (both rates are continuously compounded).
2. Enter into a forward contract to buy 1,061.84 AUD for $1,061.84 \times 0.93 = 987.51$ USD.

The 980 USD that are invested at 1% grow to

$$980e^{0.01 \times 2} = 999.80 \text{ USD}$$

Of this, 987.51 USD are used to purchase 1,061.84 AUD under the terms of the forward contract. This is exactly enough to repay principal and interest on the 1,000 AUD that are borrowed. The strategy therefore gives rise to a riskless profit of $999.8 - 987.51 = 12.29$ USD.

Commodity Futures

$$F_0 = S_0 e^{(r+u)T}$$

where u is the storage cost per unit time as a percent of the asset value.

Alternatively,

$$F_0 = (S_0 + U) e^{rT}$$

where U is the present value of the storage costs.



Consider a 1-year futures contract on an investment asset that provides no income. It costs \$2 per unit to store the asset, with the payment being made at the end of the year. Assume that the spot price is \$450 per unit and the risk-free rate is 7% per annum for all maturities. Determine the futures price.

Solution

Here, $r = 7\%$, $S_0=450$, $T=1$ year

Thus,

$$U = 2e^{-0.07 \times 1} = 1.865$$

$$F_0 = (450 + 1.865)e^{0.07 \times 1} = \$484.63$$

$F_{\text{actual}} > F_{\text{fair value}}$
Cash-and-carry arbitrage

On trading day (start day)

(i) Borrow amount equal to $(S+U)$ at risk free rate of interest

(ii) Buy one unit of commodity at spot price and pay storage cost

(iii) Short futures contract on one unit of commodity

On maturity day of futures

(iv) Deliver commodity and receive payment

(v) Repay borrowed amount with interest

$F_{\text{actual}} < F_{\text{fair value}}$
Reverse cash-and-carry arbitrage

On trading day (start day)

(i) Sell one unit of commodity at spot price

(ii) Invest/lend sale proceeds at risk free interest rate

(iii) Long futures contract on one unit of commodity

On maturity day of futures

(iv) Receive investment amount with interest

(v) Take delivery of commodity and make payment

Question- I

A one-year long forward contract on a non-dividend-paying stock is entered into when the stock price is \$40 and the risk-free rate of interest is 5% per annum with continuous compounding.

- a) What are the forward price and the initial value of the forward contract?*
- b) Six months later, the price of the stock is \$45 and the risk-free interest rate is still 5%. What are the forward price and the value of the forward contract?*

Question-2

The spot price of silver is \$25 per ounce. The storage costs are \$0.24 per ounce per year payable quarterly in advance. Assuming that interest rates are 5% per annum for all maturities, calculate the futures price of silver for delivery in nine months.

Question-3

Suppose that the risk-free interest rate is 6% per annum with continuous compounding and that the dividend yield on a stock index is 4% per annum. The index is standing at 400, and the futures price for a contract deliverable in four months is 405. What arbitrage opportunities does this create?

Question-4

The two-month interest rates in Switzerland and the United States are, respectively, 1% and 2% per annum with continuous compounding. The spot price of the Swiss franc is \$1.0500. The futures price for a contract deliverable in two months is also \$1.0500. What arbitrage opportunities does this create?

Consumption Assets: Storage is Negative Income

$$F_0 \leq S_0 e^{(r+u)T}$$

where u is the storage cost per unit time as a percent of the asset value. Alternatively,

$$F_0 \leq (S_0 + U) e^{rT}$$

where U is the present value of the storage costs.

Convenience Yield

- The convenience yield is the benefit/return of holding (or owning) the underlying asset till expiration of a futures contract.
- In case of a commodity futures, the benefit of holding inventory of the underlying commodity may accrue from the ability to profit from temporary shortage, or the ability to keep the production process running.
- The benefits from holding the physical asset are sometimes referred to as the convenience yield provided by the commodity.
- The convenience yield reflects the market's expectations concerning the future availability of the commodity. The greater the possibility that shortages will occur, the higher the convenience yield.
- If users of the commodity have high inventories, there is very little chance of shortages in the near future and the convenience yield tends to be low.

Convenience Yield

If the dollar amount of storage costs is known and has a present value U , then the convenience yield y is defined such that

$$F_0 e^{yT} = (S_0 + U) e^{rT}$$

If the storage costs per unit are a constant proportion, u , of the spot price, then y is defined so that

$$F_0 e^{yT} = S_0 e^{(r+u)T}$$

$$F_0 = S_0 e^{(r+u-y)T}$$

The Cost of Carry

- The cost of carry, c , is the storage cost plus the interest costs less the income earned
- For an investment asset $F_0 = S_0 e^{cT}$
- For a consumption asset $F_0 \leq S_0 e^{cT}$

Futures Prices & Future Spot Prices

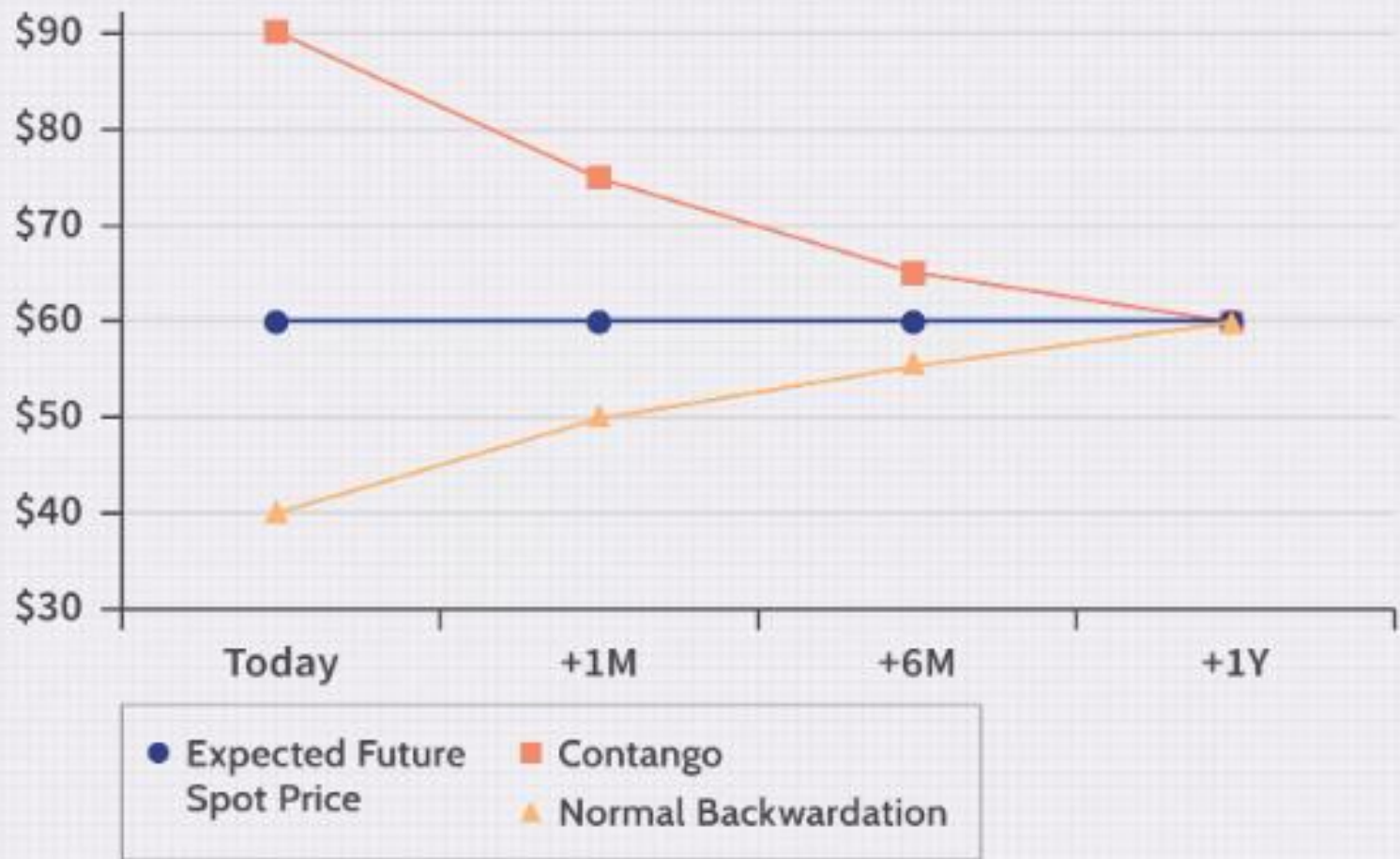
No Systematic Risk	$k = r$	$F_0 = E(S_T)$
Positive Systematic Risk	$k > r$	$F_0 < E(S_T)$
Negative Systematic Risk	$k < r$	$F_0 > E(S_T)$

Positive systematic risk: stock indices

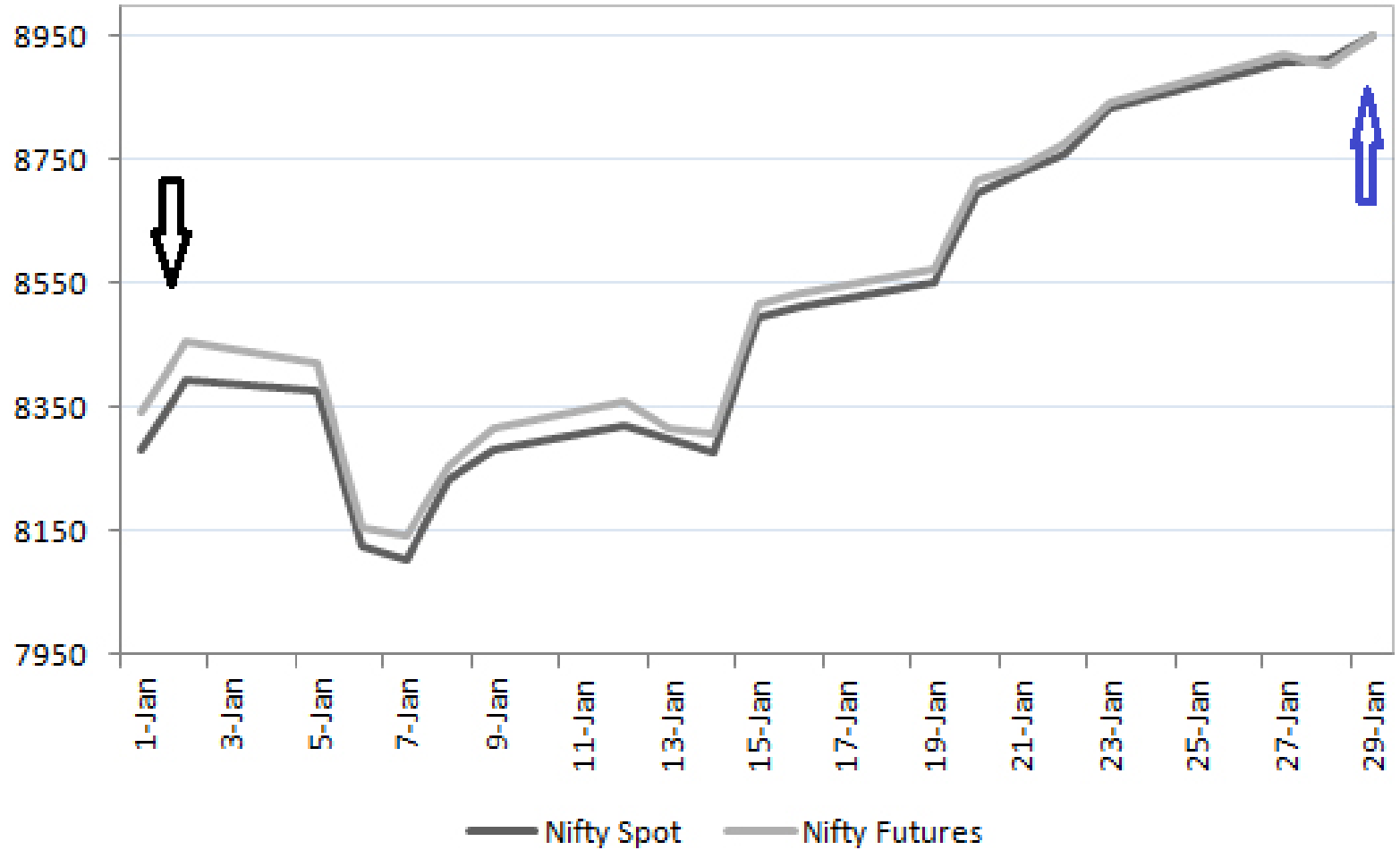
Negative systematic risk: gold (at least for some periods)

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- **Contango** is when the futures price is above the expected future spot price. Because the futures price must converge on the expected future spot price, contango implies futures prices are falling over time as new information brings them into line with the expected future spot price.
 - **Normal backwardation** is when the futures price is below the expected future spot price. This is desirable for speculators who are net long in their positions: they want the futures price to increase. So, normal backwardation is when the futures prices are increasing.

Futures Price of a Contract Due in One Year (Going Forward in Time)



Nifty Spot vs Futures





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